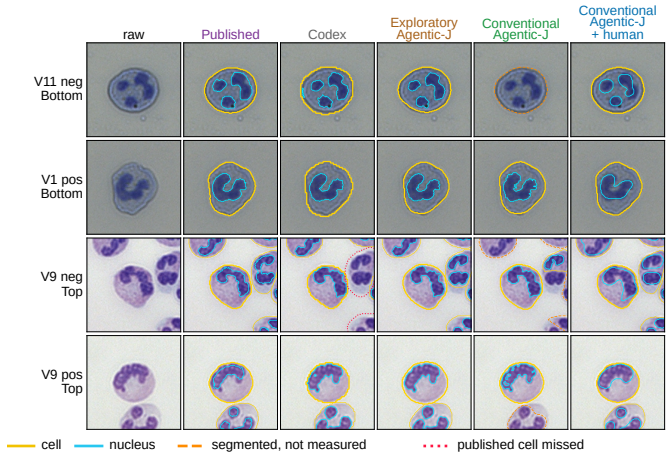
a

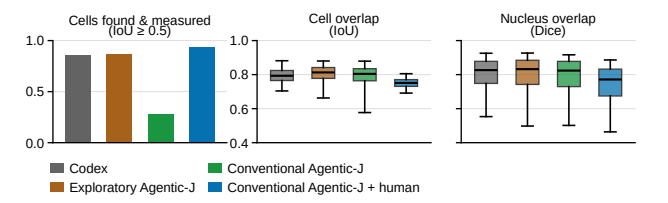
Analysis	Time	Scientific verdict	Time T1 / T2	Task 1 TUNEL calls (6 nuclei)	Task 2 published Fig. 4C (15 parasites)
		CD177 neutrophil morphology		Leishmania infection: TUNEL and proliferation	
Published	weeks	no meaningful difference	not recorded	2/6 nuclei TUNEL+	8 TUNEL+ vs 7 TUNEL−, p = 0.0012
Codex CLI	13 min	✓ 0/30 features significant	18 min (both)	✓ 6/6 calls, r = 0.98	✓ 10/15 found, hosts right; p = 0.017
Exploratory Agentic-J	35 min	✗ 5/25 significant (3/3 runs)	22 / 69 min	✓ 5/6 calls, r = 0.95, 1 split	✗ 0/15 found (+22), no host
Conventional Agentic-J	67 min	✓ 0/32 significant (3/3 runs)	55 / 85 min	✓ 5/6 calls, r = 0.99	✗ 3/15 found (+2), no TUNEL+
Conventional Agentic-J + human	488 min	✓ 0/11 features significant	32 min / n.r.	✓ 5/6 calls, r = 0.97	✓ 15/15; 14/15 within 0.1; p = 0.0012

Manual (analyst) · Human feedback (user) · Automatic (Agentic-J) · Automatic (other agent) · ✓ published conclusion reproduced · ✗ not reproduced

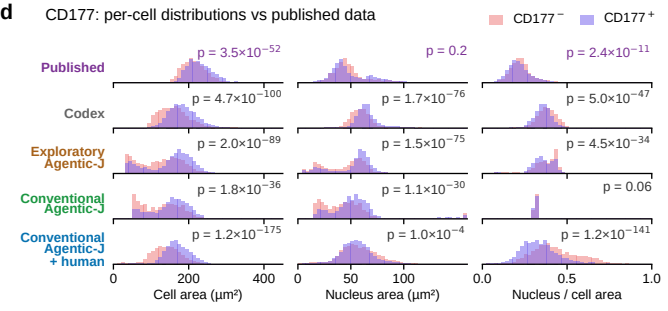
b CD177: segmentations of four example cells



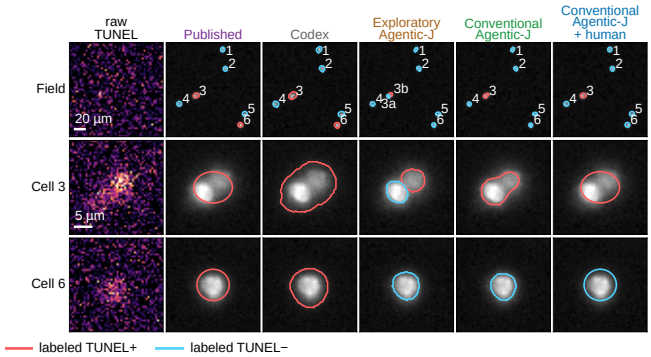
c CD177: agreement with the published masks (4389 cells)



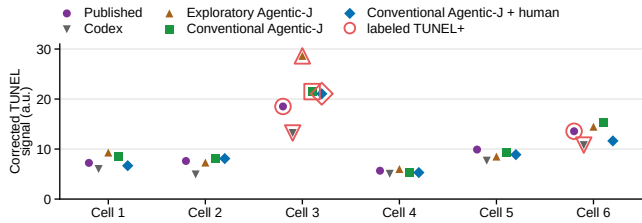
d CD177: per-cell distributions vs published data



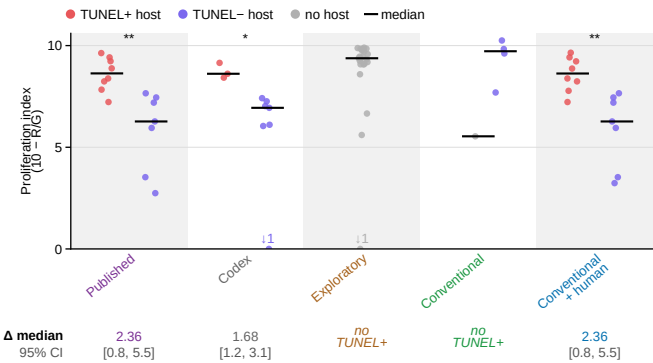
e Leishmania task 1: six nuclei, called by each pipeline



f Leishmania task 1: corrected TUNEL signal per nucleus



g Leishmania task 2: proliferation index (published Fig. 4C)



Δ median	2.36	1.68	no TUNEL+	no TUNEL+	2.36
95% CI	[0.8, 5.5]	[1.2, 3.1]			[0.8, 5.5]